

Topological Grid Segmentation - TGI Use Case Notes

[UC22] - Awesense Platform Use Case



Overview

This manual is intended to:

- Provide instructions to create the selected Use case in True Grid Intelligence (TGI).

For more details about Topological Grid Segmentation and how it can be analyzed using Awesense Platform, please refer to the [UC22 - Topological Grid Segmentation.pdf](#) document.

TGI Use Case Notes

This document provides examples of using this Use Case, a description of how to create a similar Use Case on your data, and screenshots directly from the TGI visualization interface.

Segmentations can be found directly in TGI in the main menu (select from the main menu - top hamburger icon).

The screenshot shows the True Grid Intelligence (TGI) Segmentation interface. At the top, there's a navigation bar with the "True Grid Intelligence" logo and a "Segmentations" tab highlighted with a red box. Below the navigation bar is a map of the United States with various cities and states labeled. A scale bar indicates 300 km and 200 mi. Below the map, there's a table titled "Segmentations" with columns: NAME, GRID, SEGMENTS, PUBLISHED, and CREATED. The table contains one row with the name "transformer", GRID "North Central Zone", SEGMENTS "17,632", PUBLISHED "Yes", and CREATED "25 Apr, 2023 1:50 PM". A "Create Segmentation" button is highlighted with a red box in the bottom right corner of the interface. The bottom of the interface shows a pagination bar with "Page: 1 / 1" and a small blue box with the number "1".

NAME	GRID	SEGMENTS	PUBLISHED	CREATED
transformer	North Central Zone	17,632	Yes	25 Apr, 2023 1:50 PM

To create a new segmentation, click the "Create Segmentation" button.

Create Segmentation ×

Select a grid and add segmentation rules to create a new segmentation.

Name:

Circuit to segment:

Description:

RULES

Type:

is approximately

Value:

Weight:

Add Rule

- Number of consumers
- Number of transformers
- Sum of kVA
- Line length (km)

Create Segmentation

Parameters such as “Name”, and “Circuit” (e.g. North Central Zone or awefice) must be entered in the pop-up window. “Name” serves as the name of the segmentation, “Circuit” parameter denotes the ID of the grid which should be segmented.

Next, it is necessary to select the criteria by which to segment the grid. For example, if we want to create a segment for each transformer, we enter in “RULES”, and “Type: Number of transformers” and enter 1 in “Value”. Weight is used when considering multiple criteria, so if using one criterion, enter 1. Then click on "Add Rule" and then "Create Segmentation".

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The progress of the segmentation is displayed in the upper right corner when you click on "Bell".

The screenshot shows the True Grid Intelligence Segmentations page. On the left, the 'Grid Element' details for 'arc:PRIMARY_CONDUCTOR' are displayed. The 'Trace' section shows various filters like Source, Down, Same Voltage, Connected, Nearby One Way, Nearby Other Way, Nearby Both Ways, and All Sources. The 'Segmentations' table lists the following data:

NAME	GRID	SEGMENTS	PUBLISHED	CREATED
transformer	North Central Zone	17,632	Yes	25 Apr, 2023 1:50 PM

An 'Activity Monitor' pop-up is visible in the top right corner, showing the progress of the segmentation process. It indicates that the process is 'Segmenting grid awefice' and is 51% complete. The progress bar shows a green line representing the progress.

Once the segmentation process has finished, the new segmentation will be displayed in the main menu on the Segmentations page. Individual segments can be selected and explored on the map or by clicking on "Base Segments" at the bottom left.

The screenshot shows the True Grid Intelligence Segmentations page after the segmentation process has completed. The 'test' segmentation is highlighted, showing 6 segments. The 'Segmentations' table lists the following data:

NAME	GRID	SEGMENTS
test	awefice	6
transformer	North Central Zone	17,632

The 'Base Segments' button is highlighted at the bottom left. The map shows the segmented grid elements, with the 'test' segmentation highlighted in blue. The 'Segmentations' table is also highlighted with a red box.

Once the user is satisfied with the segmentation, the user can click “Publish Segmentation” and the segmentation instance will become available throughout the Awesense Platform.

General Note

- This description was created for TGI version 3.10.0. For future versions of TGI, the process might be slightly different.
- Access to the different parts of TGI varies depending on the permissions of the user.
- Users on the Sandbox environments currently have read-only permissions, therefore, this use case may be accessible only for viewing, not creation or editing. If you are interested in the latter, please contact Awesense.
- Some of the data for this use case may be available only on Sandbox tiers 2 & 3, but not on tier 1. If you are interested in upgrading your tier access, please contact Awesense.
- For more information please contact tools@awesense.com.